

Dashrath Paswan

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Profile

GCP Data Engineer with 2 years of experience in building and optimizing large-scale pipelines on Google Cloud. Expert in **Apache Airflow** and high-concurrency Python (**Asyncio**, **Multithreading**) for workflow orchestration. Strong foundation in **Data Structures, Algorithms, and problem-solving**, with 2000+ DSA problems solved (LeetCode Knight). Hands-on experience with scalable system design (HLD & LLD), cloud-native data engineering, and Generative AI (RAG) applications.

Experience

Tata Consultancy Services (TCS) — GCP Data Engineer

Aug 2024 – Present

- **Performance Engineering:** Engineered a high-concurrency data ingestion pipeline utilizing **Asyncio** for asynchronous API fetches and **Multithreading** for BigQuery loading, reducing Apache Airflow DAG runtime from **6-7 hours to 15 minutes** (25x+ speedup).
- Built pipelines with **Apache Beam (Java and Python)** on Dataflow, processing 5TB+ daily.
- Optimized **BigQuery** with partitioning, cutting costs by 35% and improving SLA from 90% to 99.9%.

Amazon — Software Development Engineer Intern

May 2023 – Jul 2023

- Designed and implemented **end-to-end (E2E) and unit test suites** for the Artwork Library platform using **Cypress**, ensuring functional and regression coverage.
- Automated validation workflows by integrating with **CI/CD pipelines (AWS, Git)**, reducing manual QA effort by **40%** and increasing release reliability.
- Collaborated with cross-functional teams to streamline testing, bug reporting, and release processes, improving delivery efficiency.

Education

Indian Institute of Technology Patna — B.Tech, Electrical and Electronics Engineering

Jul 2020 – May 2024

Projects

JARVIS: Autonomous AI Recruitment Agent — Python, Playwright, LangChain, LLMs, SQLite

GitHub

- Developed a multi-modal AI agent using **Playwright** and **LangChain** to autonomously navigate dynamic web UIs and submit high-volume job applications.
- Integrated **GPT-OSS-120B** to dynamically resolve complex, multi-step recruiter questionnaires with context-aware responses.
- Implemented a persistent **SQLite-based LLM cache** to store recurring query responses, reducing API latency and optimizing token usage.
- Engineered human-mimicry algorithms featuring stochastic delays, variable-speed typing, and randomized mouse trajectories to bypass anti-bot systems.

Retail Stock Forecasting (TCS) — GCP, Airflow, Python, SQL, AWS, PyTest

- Built an **end-to-end ETL pipeline** using Airflow (Cloud Composer) to prepare ML-ready datasets for stock demand forecasting.
- Extracted, cleaned, and loaded production retail data into **BigQuery** with unit testing and data quality checks.
- Securely transferred curated datasets from **GCP to AWS** using IAM Anywhere role in required file formats for cross-cloud consumption.

Skills

Cloud & Data Engineering: GCP (Dataflow, BigQuery, BigQuery ML, GCS, Cloud Composer, Pub/Sub, IAM), Apache Airflow, Data Lakes, REST API, FastAPI

Generative AI & LLMs: RAG, LangChain, Ollama, ChromaDB (Vector Database), Streamlit, Prompt Engineering

Languages & Core Python: Python (**Asyncio**, **Multithreading**, **Concurrency**), Java, C, C++, SQL

Frameworks/Tools: Apache Beam, Docker, Git, Jenkins, Bitbucket, Jira, Cypress, Postman

Databases: MongoDB, MySQL, BigQuery, SQLite

Concepts: System Design (HLD & LLD), Data Structures & Algorithms, OOP, DBMS, OS, Computer Networks

Accomplishments

- Achieved **Knight rank** (1925) on LeetCode — Profile
- Achieved **Specialist rank** (1409) on Codeforces

Certifications

- **Professional Data Engineer** — Google Cloud (Credly)
- **Professional Machine Learning Engineer** — Google Cloud (Credly)